



W1-2-60-1-6

**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE AND TECHNOLOGY**

University Examinations 2022/2023

**SECOND YEAR SECOND SEMESTER EXAMINATION FOR THE
DEGREE OF BACHELOR OF SCIENCE IN GEOMATIC ENGINEERING
AND GEOSPATIAL INFORMATION SYSTEMS / SECOND YEAR
SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN GEOSPATIAL INFORMATION SCIENCE
EGE 2207 / EGS 2209: CARTOGRAPHY II / CARTOGRAPHY &
VISUALISATION II**

DATE: APRIL 2023

TIME: 2 HRS

INSTRUCTIONS: ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS

QUESTION ONE (30 MARKS) - COMPULSORY

- a) Briefly describe the CAD basic structure (5 marks)
- b) What are the data requirements for an effective data classification scheme (5 marks)
- c) Discuss data suitability and legend designs for proportional symbol maps (5 marks)
- d) Explain different color theories (5 marks)
- e) Briefly describe some of dashboard types applicable for online visualization (5 marks)
- f) How are data measurement levels applied in map symbology design (5 marks)

QUESTION TWO (20 MARKS)

- a) Discuss the basis of isarithmic thematic map construction (7 marks)
- b) Differentiate the main types of graphic technology applied in CAD (6 marks)
- c) Explain some of the guidelines followed when labeling features on a digital topographical map (7 marks)

QUESTION THREE (20 MARKS)

- a) Describe planar and hierarchical organization as used in map composition (7 marks)
- b) Distinguish between color schemes and models (7 marks)
- c) Write short notes on examples of CAD transformation operations (6 marks)

QUESTION FOUR (20 MARKS)

- a) Describe types of layout designs used when designing web maps (6 marks)
- b) Discuss the modern CAD design process (7 marks)
- c) Briefly discuss the map reproduction process (7 marks)

QUESTION FIVE (20 MARKS)

- a) What are some of considerations when designing dot density thematic maps (7 marks)
- b) Using suitable illustrations explain the types of dynamic maps (6 marks)
- c) Distinguish between wireframe and surface modeling CAD operations (7 marks)